RV College of Engineering®

MACHINE LEARNING OPERATIONS

Unit 5:

Who Decides What Governance an Organization Needs?

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PRESENTATION OVERVIEW

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CHAPTER 01

Introduction:
The Critical Role of Governance

The Governance Imperative

As organizations increasingly deploy machine learning (ML) models to automate decisions, governance becomes a **critical requirement** rather than an optional control. Governance in MLOps refers to the policies, processes, roles, and controls that ensure machine learning systems are **reliable, compliant, ethical, and aligned with business objectives**.

CENTRAL QUESTION

Who decides what level of governance an organization needs?

This presentation answers this fundamental question in MLOps governance.

- **Collective Decision-Making**

  Governance decisions are not made by a single individual or team, but are collectively determined.

- **Shaped by Multiple Forces**

  Risk, regulation, organizational structure, and business impact collectively shape governance.

Reliability

Ensuring consistent and dependable model performance

Compliance

Adhering to laws, regulations, and standards

Business Alignment

Supporting organizational objectives

CHAPTER 02

What Governance Means in MLOps

In the context of MLOps, governance encompasses the **policies, processes, roles, and controls** that ensure machine learning systems are reliable, compliant, ethical, and aligned with business objectives. It exists to **reduce model risk**, which includes incorrect predictions, bias, data misuse, security vulnerabilities, and operational failures.

01

Rules for Model Lifecycle

Defining standardized rules, best practices, and quality standards for model development, deployment, and monitoring.

Development · Deployment · Monitoring

02

Approval Workflows

Establishing clear approval and validation workflows that ensure models are properly vetted before deployment.

Validation · Approval · Workflows

03

Accountability

Ensuring accountability and auditability of models by maintaining clear ownership and traceability.

Ownership · Traceability · Audit Trails

04

Compliance

Managing compliance with laws, regulations, industry standards, and ethical guidelines.

Legal · Ethical · Standards

05

Access & Versioning

Controlling access, versioning, and lifecycle management of models and data.

Access Control · Versioning · Lifecycle

CHAPTER 03

Governance Is Not Decided by a Single Role

A Fundamental Principle

A key principle of MLOps governance is that **no single team can define governance requirements in isolation**. This is because machine learning systems affect multiple dimensions of an organization simultaneously, creating interconnected risks and responsibilities that require diverse perspectives and expertise.

Why Governance Cannot Be Siloed

When organizations allow a single team to define governance, they create blind spots. Technical teams may underestimate business impact, business teams may overlook regulatory requirements, and compliance teams may not fully understand model complexity. Effective governance requires **cross-functional collaboration** from the outset.

Collective Approach

Multiple stakeholders contribute their unique expertise to define comprehensive governance

Siloed Approach

Single teams create blind spots, leading to inadequate or misaligned governance

Business Outcomes

ML models directly impact revenue, customer experience, and competitive advantage. Governance must align with business objectives and risk tolerance.

Technical Systems

Models introduce technical complexity, infrastructure requirements, and system dependencies that affect performance and reliability.

Legal & Regulatory

Automated decisions may violate regulations, creating legal liability and financial penalties if not properly governed.

Ethical & Societal

Models can perpetuate bias, discriminate unfairly, and impact society in ways that damage reputation and trust.

Primary Stakeholders in Governance Decisions

Executive Leadership

Strategic Governance

Senior leadership plays a foundational role in governance decisions by setting organizational risk tolerance and strategic priorities.

Responsibilities:

- Defining organizational risk tolerance
- Setting strategic priorities for AI/ML usage
- Deciding acceptable control and oversight levels

KEY QUESTION

What is the potential business impact if this model fails, behaves incorrectly, or causes harm?

Business Owners & SMEs

Business Impact Assessment

Business teams and domain experts define how models are used in real-world processes and assess their practical impact.

Responsibilities:

- Clarifying business objectives and decision contexts
- Assessing impact of incorrect or biased predictions
- Providing feedback on model performance

KEY QUESTION

How critical is this model's decision to customers, users, or revenue?

Primary Stakeholders in Governance Decisions

Data Science & Engineering

Technical Risk Management

Technical teams understand the internal behavior, limitations, and risks of ML models from a technical perspective.

Responsibilities:

- Identifying technical risks (bias, drift, data leakage)
- Ensuring reproducibility and traceability
- Implementing monitoring and validation

KEY QUESTION

How complex, fragile, or opaque is this model technically?

Risk, Compliance & Audit

Regulatory & Legal Oversight

Especially critical in regulated industries (finance, healthcare, pharmaceuticals), these teams ensure legal compliance.

Responsibilities:

- Ensuring compliance with policies and regulations
- Validating models before deployment
- Maintaining audit trails and documentation

KEY QUESTION

What governance controls are legally or regulatorily required?

Governance Is Driven by Risk Level

A central concept in MLOps governance is **risk-based governance**. Governance requirements should **scale with the level of risk** posed by a model. This approach ensures that resources and controls are appropriately allocated, avoiding both under-governance (exposing the organization to undue risk) and unnecessary bureaucracy (slowing innovation).

Low-Risk Use Cases

Examples:

- Internal analytics and reporting
- Decision-support tools with human oversight
- Limited impact of individual errors

Governance Characteristics:

- ✓ Lightweight approvals
- ✓ Basic monitoring
- ✓ Minimal documentation
- ✓ Standard security

High-Risk Use Cases

Examples:

- Customer-facing decisions (credit, pricing)
- Financial, medical, or legal outcomes
- Fully automated decision-making

Governance Characteristics:

- ✓ Formal validation workflows
- ✓ Continuous monitoring
- ✓ Human oversight paths
- ✓ Detailed auditability

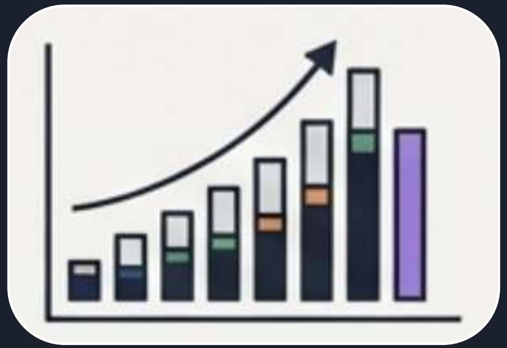
The Risk-Matching Approach

This approach avoids both under-governance (exposing to risk) and over-governance (slowing innovation).

Governance Is Driven by Risk Level

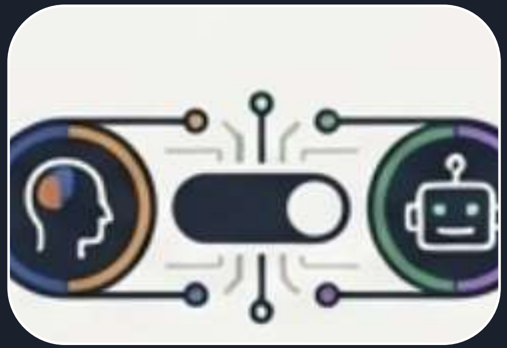
Risk level is neither deterministic nor categorical; rather, it exists on a spectrum.

And these 6 factors determine the model's position on the risk spectrum:



1. Severity of Impact
What are the consequences if predictions are wrong? (Financial, legal, personal)

2. Scale of Deployment
How many users or decisions are affected?



3. Degree of Automation
Is there a human-in-the-loop, or is the decision fully automated?

4. Regulatory Environment
Is the model operating in a regulated industry like finance or healthcare?

5. Ethical Implications
Does the model touch on sensitive areas like fairness, bias, or privacy?

6. Public Visibility
What is the reputational risk if the model's behavior becomes public?

A RACI matrix translates shared responsibility into a clear operational playbook.

This matrix clarifies who is responsible, accountable, consulted and informed for each stage of the MLOps lifecycle it provides a starting point for defining roles and ensuring nothing falls through the cracks

Tasks	Business Stakeholders	Business Analysis/ Citizen DS	Data Scientists	Risk/ Audit	Data Ops	Production/ Exploitation	Resources/ Architect
Identification	A/R	C		I			
Data Preparation	C	A/R	C				
Data Modeling	C	A	R				
Model Acceptance	I	C	C	A/R			
Productionalization		C	A/R	I	C		
Capitalization			R		R		A
Integration to external systems					A/R		
Global Orchestration		C			R	A	
User acceptance tests	A/R	R	C	I			
Deployments					R	A	I
Monitoring	I	C				A/R	

A: Accountable, R: Responsible, C: Consulted, I: Informed

Factors That Influence Governance Requirements

> Several factors determine how strict governance must be. The higher these factors score, the **stronger governance controls** are required to mitigate risk and ensure responsible deployment of ML systems.

01

Severity of Impact

How severe are the consequences if the model's predictions are wrong? This includes financial loss, harm to individuals, legal liability, and reputational damage.

Higher impact = Stronger controls

02

Scale of Deployment

How many users or decisions are affected by the model? A model affecting millions of customers requires stronger governance than one used by a small team.

Larger scale = Stronger controls

03

Degree of Automation

Is the decision fully automated or does a human review and approve each outcome? Fully automated decisions require stronger governance.

More automation = Stronger controls

Factors That Influence Governance Requirements

> Several factors determine how strict governance must be. The higher these factors score, the **stronger governance controls** are required to mitigate risk and ensure responsible deployment of ML systems.

04
Regulatory Environment

What laws and regulations apply to the model's decisions? Industries like finance and healthcare have strict requirements demanding stronger governance.

More regulated = Stronger controls

05
Ethical & Societal Impact

Could the model perpetuate bias, discriminate unfairly, or cause societal harm? Models with ethical implications need stronger oversight.

Higher ethical risk = Stronger controls

06
Public Visibility

How visible is the model to regulators, media, and the public? High-visibility models carry reputational risk requiring stronger governance.

Higher visibility = Stronger controls

CHAPTER 07

Governance Is a Continuous Process

Beyond One-Time Design

Governance is **not a one-time design decision** that remains static. It must evolve continuously as the organization, its models, and the external environment change. Static governance quickly becomes outdated and ineffective.

Why Governance Must Evolve

Machine learning systems are dynamic. Models are retrained, deployed in new contexts, and used in ways that weren't originally anticipated. Data drifts, regulations change, and new risks emerge. Governance that doesn't evolve with these changes will fail to protect the organization.

∞ Continuous Governance Cycle

Effective organizations establish a regular cadence for reviewing and refining governance policies, ensuring they remain aligned with real-world usage and emerging risks.

Model Reuse

Models are repurposed for new use cases, potentially introducing unforeseen risks.

Data Drift

Data distributions change over time, affecting model performance and reliability.

Regulatory Updates

New laws and regulations emerge, changing compliance requirements.

System Scaling

Systems scale across teams and regions, amplifying both benefits and risks.

Incident Revelation

Failures and near-misses reveal new risks and vulnerabilities.

CONCLUSION

Principles for Effective Governance

01 Collective Decision-Making

Governance decisions are collective, requiring input from executives, business leaders, technical teams, and risk managers. No single role can define appropriate governance in isolation.

02 Risk-Based Approach

Governance should match the risk level of the model. Low-risk models need lightweight controls, while high-risk models require comprehensive oversight and validation.

03 Enables Responsible Scaling

Strong governance does not slow innovation—it enables sustainable, trustworthy, and accountable use of machine learning at scale.

04 Protects Organization & Users

Proper governance protects both the organization and its users from harm, ensuring ML systems are reliable, fair, and beneficial.

Good MLOps governance does not slow innovation

It enables **sustainable, trustworthy, and accountable** use of machine learning

Thank You